

CAPITAL STRUCTURE

1. Capital Structure

In this company has to decide the optimal mix of capital.

2. Objectives of FM

Wealth Maximization i.e. Dividend & Capital appreciation

3. $PE\ Ratio = \frac{MPS}{EPS}$ ✓

$MPS = PE\ Ratio \times EPS$

4. $PE\ Ratio = \frac{1}{Ke}$ or $Ke = \frac{1}{PE\ Ratio}$ ✓

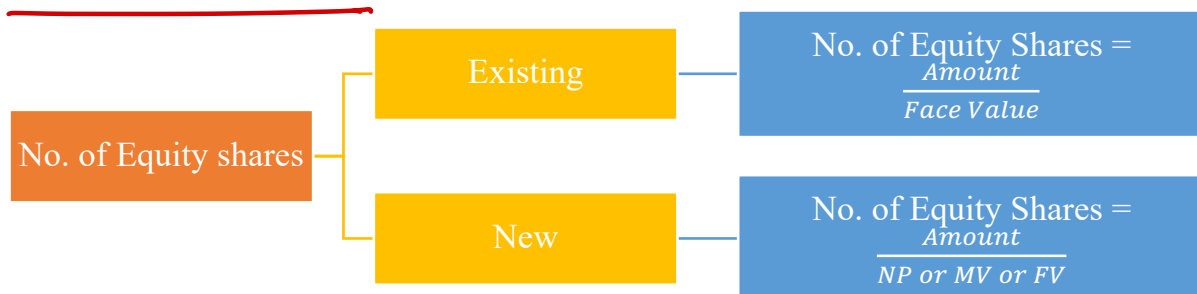
5. Statement of MPS

Particulars	Plan – A	Plan – B
EBIT (Operating Profit)		
(-) Interest		
EBT		
(-) Tax		
EAT		
(-) Preference Dividend		
EAFE (Earning available for Equity)		
No. of Equity Shares		
EPS		
PE Ratio		
MPS		

6. Interest and Preference Dividend



7. Number of Equity Shares



8. Points to Remember (PTRs)

- EBIT will remain same for all options
- EBIT is independent of capital structure
- EBIT is dependent on amount of capital employed
- Rate of return on capital employed will remain same, unless and until specifically mentioned in question
- New EBIT = New capital employed × Return on capital employed (ROCE)
- Return on capital employed (ROCE) = $\frac{\text{Existing EBIT}}{\text{Existing Capital Employed}}$ ✓
- In case if MPS can't be computed than decision will be based on EPS.

Question – 1

The particulars relating to Raj Ltd. for the year ended 31st March, 2022 are given as follows:

Output (units at normal capacity)	→	1,00,000
Selling price per unit	→	₹ 40
Variable cost per unit	→	₹ 20
Fixed cost	→	₹ 10,00,000

$$\text{Ex. } \text{EBIT} = (40 - 20)(100) - 100 = 100$$

$$\text{New EBIT} = (40 - 17)(1.500) - 150 = 19.500$$

The capital structure of a company as on 31st March, 2022 is as follows:

Particulars	Amount in ₹
Equity share capital (1,00,000 shares of ₹ 10 each)	→ 10,00,000
Reserve and surplus	→ 5,00,000
Current liabilities	→ 5,00,000
Total:	20,00,000

Raj Ltd. has decided to undertake an expansion project to use the market potential that will involve ₹ 20 lakhs. The company expects an increase in output by 50%. Fixed cost will be increase by ₹ 5,00,000 and variable cost per unit will be decreased by 15%. The additional output can be sold at existing selling price without any adverse impact on the market.

The following alternative schemes for financing the proposed expansion program are planned:

Alternative	Debt	Int	Equity Shares	Balance	New No. of Sh.	Total Sh.
1 $(50 \times 10\%)$	5,00,000	50000	150 =	Balance ÷ 200 = 7500	7500	107500
2 $(50 \times 10\%) + (50 \times 15\%)$	10,00,000	125000	100 =	Balance ÷ 200 = 5000	5000	105000
3 $(50 \times 10\%) + (50 \times 15\%) + (100 \times 20\%)$	14,00,000	205000	60 =	Balance ÷ 200 = 3000	3000	103000

Current market price per share is ₹ 200.

Slab wise interest rate for fund borrowed is as follows:

Fund Limit	Applicable interest rate
Up-to ₹ 5,00,000	→ 10%
Over ₹ 5,00,000 and up-to ₹ 10,00,000	→ 15%
Over ₹ 10,00,000	→ 20%

Find out which of the above-mentioned alternatives would you recommend for Raj Ltd. with reference to the EPS, assuming a corporate tax rate is 40%?

Solution**Calculation of EBIT**

Particulars	Existing	Proposed
Sale units	1,00,000	1,50,000
Contribution per unit	40 - 20 = 20	40 - (20 × 85%) = 23
Total contribution	20,00,000	34,50,000
Less: Fixed cost	10,00,000	15,00,000
EBIT	10,00,000	19,50,000

Statement of EPS

Particulars	Existing	Alternative – 1	Alternative – 2	Alternative – 3
EBIT	→ 10,00,000	→ 19,50,000	→ 19,50,000	→ 19,50,000
Less: Interest	→ -	✓ 50,000 (5,00,000 × 10%)	✓ 1,25,000 [(5lakh×10%) + (5lakh×15%)]	2,05,000 [(5lakh×10%) + (5lakh×15%) + (4lakh×20%)]
EBT	→ 10,00,000	19,00,000	18,25,000	16,95,000
Less: Tax @ 40%	→ 4,00,000	7,60,000	7,30,000	6,78,000
EAT / EAE (A)	→ 6,00,000	11,40,000	10,95,000	10,17,000
No. of Equity Shares				
- Existing	1,00,000	1,00,000	1,00,000	1,00,000
- New	-	$\frac{15,00,000}{200} = 7,500$	$\frac{10,00,000}{200} = 5,000$	$\frac{6,00,000}{200} = 3,000$
Total Equity Shares (B)	1,00,000	1,07,500	1,05,000	1,03,000
EPS (A ÷ B)	6.00	10.60	10.43	9.87

Since, Alternative – 1 has highest EPS, thus it is recommended to raise funds in combination of debt of ₹ 5,00,000 and balance ₹ 15,00,000 from equity.

Ex. Cap. = (0.80 × 10) + 12 + 10 = 30
 $\text{New EBIT} = \frac{4.50}{30} \times 36 = 5.40$

Question – 2

Earnings before interest and tax of a company are ₹ 4,50,000. Currently the company has 80,000 Equity shares of ₹ 10 each, retained earnings of ₹ 12,00,000. It pays annual interest of ₹ 1,20,000 on 12% Debentures. The company proposes to take up an expansion scheme for which it needs additional fund of ₹ 6,00,000. It is anticipated that after expansion, the company will be able to achieve the same return on investment as at present. It can raise fund either through debts at rate of 12% p.a. or by issuing Equity shares at par. Tax rate is 40%.

→ $60 \div 10 = 6000 \text{ sh.}$

→ $60 \times 12\% = 0.72$

Required to compute the earning per share if:

- The additional funds were raised through debts.
- The additional funds were raised by issue of Equity shares.

Advise whether the company should go for expansion plan and which sources of finance should be preferred.

Solution

Existing capital employed = Equity + Retained Earnings + Debentures

= (80,000 × 10) + 12,00,000 + (1,20,000 ÷ 12%) = ₹ 30,00,000

Capital employed after expansion = 30,00,000 + 6,00,000 = ₹ 36,00,000

New EBIT = $\frac{\text{Existing EBIT}}{\text{Existing Capital}} \times \text{New Capital} = \frac{4,50,000}{30,00,000} \times 36,00,000 = ₹ 5,40,000$

Statement of EPS

Particulars	Existing	Additional fund as debt	Additional fund as equity
EBIT	→ 4,50,000	→ 5,40,000	→ 5,40,000
Less: Interest			

- Existing Debt	→	✓ 1,20,000	✓ 1,20,000	✓ 1,20,000
- <u>New Debt</u>		-	✓ 72,000	✓ -
EBT	→	3,30,000	3,48,000	4,20,000
Less: Tax @ 40%	→	1,32,000	1,39,200	1,68,000
EAT/EAE (A)	→	1,98,000	2,08,800	2,52,000
No. of Equity shares (B)	→	✓ 80,000	✓ 80,000	80,000 + 60,000 = 1,40,000
EPS (A ÷ B)		2.475	2.610	1.800

EPS is higher when the additional funds are raised through debt, thus it is the recommended option for the company.

$$\text{Cap. Cost} = (30000 \times 10\%) + 72 + 48 = 140$$

$$\text{New EPS} = \frac{2.808}{140} \times 182 = 3.602$$

$$\text{Debt} - \Delta P = \frac{82 \times 100}{182} = 44.44\%$$

$$\rightarrow P/E = 8 + 2.4 \times 12\%$$

Question – 3

Akash Limited provides you the following information:

	Amount (₹)
Profit (EBIT)	→ 2,80,000
Less: Interest on Debentures @ 10%	→ (40,000)
EBT	→ 2,40,000
Less: Income Tax @ 50%	→ (1,20,000)
	→ 1,20,000
No. of Equity shares (₹ 10 each)	→ ✓ 30,000
Earnings per share (EPS)	→ 4
Price /EPS – (PE Ratio)	→ ✓ 10

The company has reserves and surplus of ₹ 7,00,000 and required ₹ 4,00,000 further for modernization. Return on Capital Employed (ROCE) is constant. Debt (Debt/ Debt + Equity) Ratio higher than 40% will bring the P/E Ratio down to 8 and increase the interest rate on additional debts to 12%. You are required to ASCERTAIN the probable price of the share.

(a) if the additional capital is raised as debt; and

(b) if the amount is raised by issuing equity shares at ruling market price. → $4 \times 10 = 40$
 $\rightarrow 40 \div 40 = 10000$

Solution

Statement of MPS

Particulars	Debt Option (₹)	Equity Option (₹)
EBIT	→ 3,60,000	→ 3,60,000
Less: Interest on old deb.	→ 40,000	→ 40,000
Less: Interest on new debt (4,00,000 × 12%)	→ 48,000	○
EBT	→ 2,72,000	3,20,000
Less: Taxes @ 50%	→ 1,36,000	1,60,000
EAT/EAE	→ 1,36,000	1,60,000
Number of equity shares	✓ 30,000	30,000 + $(\frac{40}{4})$ = 40,000
Earning per share (EPS)	→ 4.53	→ 4.00
PE Ratio	○ 8	○ 10
Market Price Per Share (EPS × PE Ratio)	36.24	40

Option II of raising funds with equity is better.

Working Note:

- 1) Existing capital = Equity + 10% Debentures + Reserve & Surplus
 $= (30,000 \times 10) + (40,000 \div 10\%) + 7,00,000 = ₹ 14,00,000$
 Rate of present earnings $= \frac{2,80,000}{14,00,000} \times 100 = 20\%$
 New capital employed $= 14,00,000 + 4,00,000 = ₹ 18,00,000$
 New EBIT $= 18,00,000 \times 20\% = ₹ 3,60,000$

2) **Option I**

$$\text{Debt Equity Ratio} = \frac{\text{Debt}}{\text{Debt} + \text{Equity}} = \frac{4,00,000 + 4,00,000}{18,00,000} \times 100 = 44.44\%$$

Since Debt Equity Ratio is more than 40%, thus PE ratio will be down to 8.

Option II

$$\text{Debt Equity Ratio} = \frac{\text{Debt}}{\text{Debt} + \text{Equity}} = \frac{4,00,000}{18,00,000} \times 100 = 22.22\%$$

Since Debt Equity Ratio is less than 40%, thus PE ratio will remain same at 10.

9. Indifference Level

Level of EBIT where EPS of the two options will be equal.

$$\checkmark \text{EPS} = \frac{(\text{EBIT} - \text{Interest})(1-t) - \text{Preference dividend}}{\text{No. of equity shares}}$$

$$\begin{aligned} \text{EPS of option Debt + Equity} &= \text{EPS of option Debt + Equity + PSC} \\ \frac{(\text{EBIT} - \text{Interest})(1-t)}{\text{No. of equity shares}} &= \frac{(\text{EBIT} - \text{Interest})(1-t) - \text{Preference dividend}}{\text{No. of equity shares}} \end{aligned}$$

Solve to get EBIT

10. Decision criteria

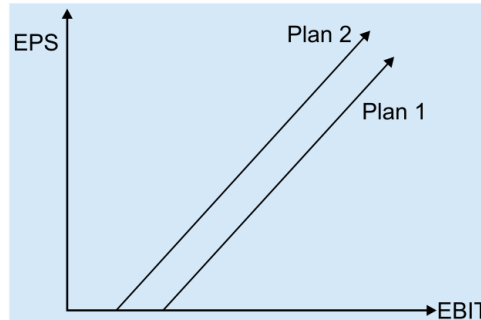
Situation	Decision
Expected EBIT < Indifference Level	Select option having lower fixed financial burden
Expected EBIT = Indifference Level	Select any option
Expected EBIT > Indifference Level	Select option having higher fixed financial burden

11. In case if there are three options then solve as follows:

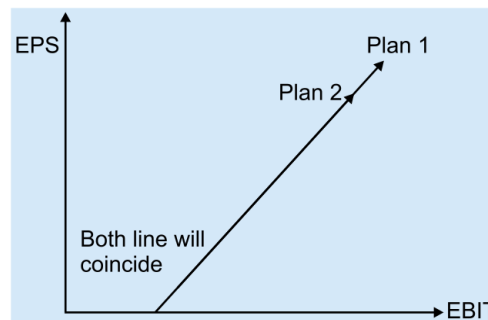
- (i) A & B ✓
- (ii) B & C ✓
- (iii) A & C ✓

12. If amount of equity share capital is same under two financial plans, then one of the following two situations will arise:

(a) **No indifference point** – It will arise when after tax cost of the source other than equity shares is not same under both plans.



(b) **Many indifference point** – It will arise when after tax cost of the source other than equity share is same under both plans.



Question – 4

J Ltd. is considering three financial plans. The key information is as follows:

(i) Total investment to be raised ₹ 4,00,000.

(ii) Plans of Financing

Plans	Equity	Debt	Preference Shares
X	100% = 4l	-	-
Y	50% = 2l	50% = 2l	-
Z	50% = 2l	-	50% = 2l

(iii) Cost of Debt – 10%

Cost of preference shares – 10%

(iv) Tax rate is 50%

(v) Equity shares of the face value of ₹ 10 each will be issued at a premium of ₹ 10 per share

(vi) Expected EBIT is ₹ 1,00,000

You are required to compute the following for each plan:

(a) Earnings per share (EPS)

(b) Financial break-even point $\rightarrow \text{Int.} + \frac{PD}{(1-t)}$

(c) Indifference Point between the plans and indicate if any of the plans dominate.

X	Y	Z
1l	1l	1l
-	0.20l	-
1l	0.80l	1l
0.50l	0.40l	0.50l
0.50l	0.40l	0.50l
-	-	0.20l
0.50l	0.40l	0.30l
0.20l	0.10l	0.10l
2.5	4	3
-	20000	40000

$$\frac{X + Y}{(EBIT - 0)(1-0.50)} = \frac{(EBIT - 20000)(1-0.50)}{20000} \quad \left| \quad \begin{aligned} EBIT &= 2(EBIT) - 40000 \\ EBIT &= 40000 \end{aligned} \right.$$

Solution**(a) Computation of Earnings Per Share (EPS)**

Particulars	Plan X	Plan Y	Plan Z
EBIT	1,00,000	1,00,000	1,00,000
Less: Interest on debt	-	20,000	-
EBT	1,00,000	80,000	1,00,000
Less: Tax @ 50%	50,000	40,000	50,000
EAT	50,000	40,000	50,000
Less: Preference Dividend	-	-	20,000
EAE (A)	50,000	40,000	30,000
No. of equity shares (B)	20,000	10,000	10,000
EPS (A ÷ B)	2.50	4.00	3.00

(b) Computation of Financial Break-even Point

$$\text{Plan X - Financial BEP} = \text{Interest} + \frac{\text{Preference Dividend}}{(1-t)} = 0 + 0 = ₹ 0$$

$$\text{Plan Y - Financial BEP} = \text{Interest} + \frac{\text{Preference Dividend}}{(1-t)} = 20,000 + 0 = ₹ 20,000$$

$$\text{Plan Z - Financial BEP} = \text{Interest} + \frac{\text{Preference Dividend}}{(1-t)} = 0 + \frac{20,000}{(1-0.50)} = ₹ 40,000$$

(c) Indifference point**Between Plan X and Y**

$$\frac{(EBIT-0)(1-0.50)-0}{20,000} = \frac{(EBIT-20,000)(1-0.50)-0}{10,000}$$

$$\frac{0.5(EBIT)}{20,000} = \frac{0.5(EBIT-20,000)}{10,000}$$

$$EBIT = 2(EBIT) - 40,000$$

$$EBIT = ₹ 40,000$$

Between Plan Y and Z

$$\frac{(EBIT-20,000)(1-0.50)-0}{10,000} = \frac{(EBIT-0)(1-0.50)-20,000}{10,000}$$

$$\frac{0.5(EBIT-20,000)}{10,000} = \frac{0.5(EBIT)-20,000}{10,000}$$

$$0.5(EBIT) - 10,000 = 0.5(EBIT) - 20,000$$

There is **no indifference point** between Plan Y and Z.

Between Plan X and Z

$$\frac{(EBIT-0)(1-0.50)-0}{20,000} = \frac{(EBIT-0)(1-0.50)-20,000}{10,000}$$

$$\frac{0.5(EBIT)}{20,000} = \frac{0.5(EBIT)-20,000}{10,000}$$

$$0.5(EBIT) = EBIT - 40,000$$

$$EBIT = ₹ 80,000$$

Exp. EBIT < 40,000 → X
 Exp. EBIT = 40,000 → X or Y
 40,000 < Exp. EBIT < 80,000 → Y
 Exp. EBIT = 80,000 → Y
 Exp. EBIT > 80,000 → Y

40,000 < Exp. EBIT < 80,000
 Y
 X
 Y

The above indifference levels are presented in the following table:

Expected Level of EBIT	Recommended plan
Less than ₹ 40,000	Plan X
Equal to ₹ 40,000	Plan X or Plan Y
Between ₹ 40,000 to ₹ 80,000	Plan Y
More than ₹ 80,000	Plan Y

From the above table, it can be clearly observed that **Plan Y** is more dominating than other plans.

$$Alt-I \rightarrow EPS = \frac{[EBIT - (3000 \times 9\%) - (3750 \times 10\%)](1-0.34) - (32 \times 50 \times 11\%)}{(72 + \frac{1250}{20})} = \frac{(0.66)(EBIT) - 42.51}{13.25} - 16.50$$

Question – 5

Current Capital Structure of XYZ Ltd is as follows:

Equity share capital of 7 lakh shares of face value ₹ 20 each

Reserves of ₹ 10,00,000

9% bonds of ₹ 3,00,00,000

11% preference capital; 3,00,000 shares of face value ₹ 50 each

Additional funds required for XYZ Ltd are ₹ 5,00,00,000.

XYZ Ltd is evaluating the following alternatives:

(I) **Proposed alternative I:** Raise the funds via 25% equity capital and 75% debt at 10%. PE Ratio in such scenario would be 12.

(II) **Proposed alternative II:** Raise the funds via 50% equity capital and rest from 12% Preference capital. PE Ratio in such scenario would be 11.

Any new equity capital would be issued at a face value of ₹ 20 each. Any new preferential capital would be issued at a face value of ₹ 20 each. Tax rate is 34%.

Determine the indifference point under both the alternatives.

Solution

Proposed Capital Structure

Capital	Proposal I	Proposal II
Equity	→ 1,25,00,000 (5,00,00,000 × 25%)	→ 2,50,00,000 (5,00,00,000 × 50%)
Debt @10%	→ 3,75,00,000 (5,00,00,000 × 75%)	-
12% Preference shares	-	→ 2,50,00,000 (5,00,00,000 × 50%)

Total Capital Structure (including new proposal)

Capital	Proposal I	Proposal II
Equity	$(7,00,000 \times 20) + 1,25,00,000$ = 2,65,00,000	$(7,00,000 \times 20) + 2,50,00,000$ = 3,90,00,000
Reserves	→ 10,00,000	10,00,000
9% Bonds	→ 3,00,00,000	3,00,00,000
10% Debt	→ 3,75,00,000	-
11% Preference Shares	→ 1,50,00,000	-
12% Preference Shares	-	2,50,00,000
Total	11,00,00,000	11,00,00,000

Let Indifference point be ₹ y.

$$\text{EPS of Proposal I} = \frac{[y - \{(3,00,00,000 \times 9\%)\} + (3,75,00,000 \times 10\%)](1 - 0.34) - (1,50,00,000 \times 11\%)}{[7,00,000 + (1,25,00,000 \div 20)]}$$

$$= \frac{(y - 64,50,000)(0.66) - 16,50,000}{13,25,000}$$

$$\text{EPS of Proposal II} = \frac{[y - (3,00,00,000 \times 9\%)](1 - 0.34) - (2,50,00,000 \times 12\%)}{[7,00,000 + (2,50,00,000 \div 20)]} = \frac{(y - 27,00,000)(0.66) - 46,50,000}{19,50,000}$$

For calculation of indifference point,

$$\frac{(y - 64,50,000)(0.66) - 16,50,000}{13,25,000} = \frac{(y - 27,00,000)(0.66) - 46,50,000}{19,50,000}$$

$$\frac{0.66(y) - 42,57,000 - 16,50,000}{1,325} = \frac{(y)(0.66) - 17,82,000 - 46,50,000}{1,950}$$

$$\frac{0.66(y) - 59,07,000}{1,325} = \frac{(y)(0.66) - 64,32,000}{1,950}$$

$$(1,287)y - 11,51,86,50,000 = (874.50)y - 8,52,24,00,000$$

$$(412.50)y = 2,99,62,50,000$$

$$y = ₹ 72,63,636.36$$

$\text{EPS of A} = \text{EPS of B}$
 $\frac{(4,800 - (12 \times 127))}{80,000} (1 - 0.34) = \frac{(4,800)(1 - 0.34) - PD}{80,000}$
 $3,168 - 33,600 = 33,600 - PD$
 $PD = 33,600$
 $PD \% = \frac{33,600}{42} \times 100 = 8.40\%$

Question – 6

XYZ Ltd. is considering the following two alternative financing plans:

Particulars	Plan A	Plan B
Equity Shares of ₹ 10 each	8,00,000	8,00,000
12% Debentures	4,00,000	
Preference shares of ₹ 100 each		4,00,000
	12,00,000	12,00,000

The indifference point between the plans is ₹ 4,80,000. Corporate tax rate is 30%. Calculate the rate of dividend on preference shares.

Solution

At indifference level i.e. when EBIT = ₹ 4,80,000

EPS of Plan A = EPS of Plan B

$$\frac{(EBIT - \text{interest})(1 - t)}{\text{No. of equity shares}} = \frac{(EBIT - \text{interest})(1 - t) - PD}{\text{No. of equity shares}}$$

$$\frac{(4,80,000 - 48,000)(1 - 0.30)}{80,000} = \frac{(4,80,000 - 0)(1 - 0.30) - PD}{80,000}$$

$$3,02,400 = 3,36,000 - PD$$

$$PD = ₹ 33,600$$

$$\text{Rate of preference dividend} = \frac{PD}{\text{Pref. sh. capital}} \times 100 = \frac{33,600}{4,00,000} \times 100 = 8.40\%$$

Question – 7

SK Limited presently has ₹ 36,00,000 in debt outstanding bearing an interest rate of 10 percent. It wishes to finance a ₹ 40,00,000 expansion programme and is considering three alternatives; additional debt at 12 percent interest, preference shares with an 11 percent dividend, and the issue of equity shares at ₹ 16 per share. The company presently has 8,00,000 shares outstanding and is in a 40 percent tax bracket.

- (a) If earnings before interest and taxes are presently ₹ 15,00,000. Determine earnings per share for the three alternatives, assuming no immediate increase in profitability.
- (b) Analyse which alternatives do you prefer. Compute how much would EBIT need to increase before the next alternative would be best.

Sol. $\text{EBIT} < 23.76 \rightarrow \text{III}$
Sol. $\text{EBIT} \geq 23.76 \rightarrow \text{I}$

Solution

- (a) Calculation of EPS

Particulars	Alternatives		
	Alternative-I: Take additional Debt	Alternative- II: Issue 11% Preference Shares	Alternative-III: Issue further Equity Shares
	₹	₹	₹
EBIT	15,00,000	15,00,000	15,00,000
Interest on Debts:			
- on existing debt @10%	(3,60,000)	(3,60,000)	(3,60,000)
- on new debt @12%	(4,80,000)	---	---
Profit before taxes	6,60,000	11,40,000	11,40,000
Taxes @ 40%	(2,64,000)	(4,56,000)	(4,56,000)
Profit after taxes	3,96,000	6,84,000	6,84,000
Preference shares dividend	---	(4,40,000)	---
EAFE	3,96,000	2,44,000	6,84,000
Number of shares	8,00,000	8,00,000	10,50,000
Earnings per share	0.495	0.305	0.651

- (b) For the present EBIT level, equity shares are clearly preferable. EBIT would need to increase by ₹23,76,000 - ₹ 15,00,000 = ₹ 8,76,000 before an indifference point with debt is reached. One would want to be comfortably above this indifference point before a strong case for debt should be made. The lower the probability that actual EBIT will fall below the indifference point, the stronger the case that can be made for debt, all other things remain the same.

Working Note:

Calculation of indifference point between debt and equity shares -

$$\frac{\text{EBIT} - ₹8,40,000}{8,00,000} = \frac{\text{EBIT} - ₹3,60,000}{10,50,000}$$

$$\frac{\text{EBIT} - ₹8,40,000}{80} = \frac{\text{EBIT} - ₹3,60,000}{105}$$

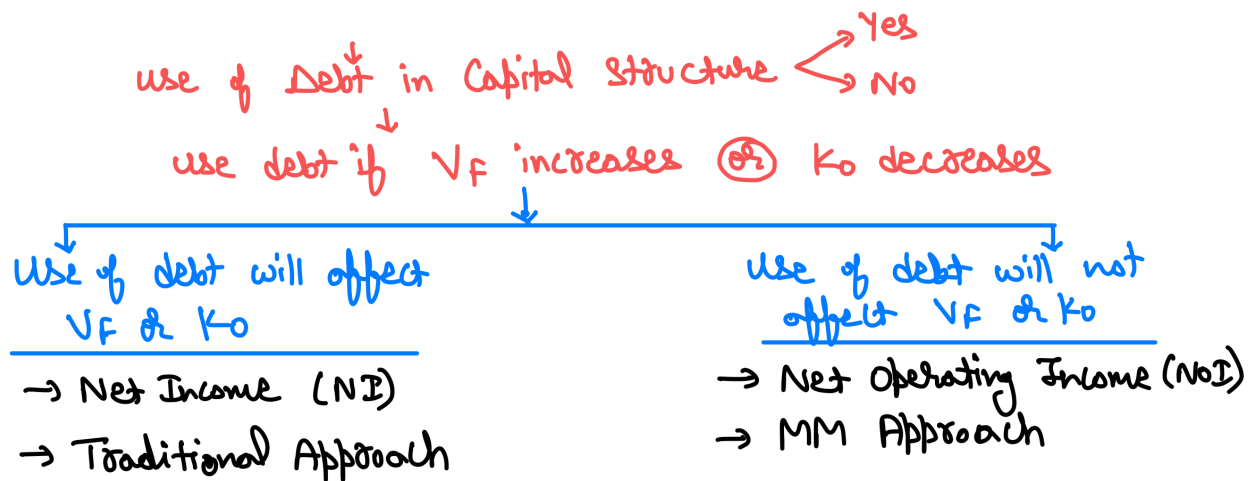
$$(\text{EBIT} - 8,40,000)(105) = (\text{EBIT} - 3,60,000)(80)$$

$$(105)\text{EBIT} - 8,82,00,000 = (80)\text{EBIT} - 2,88,00,000$$

$$(25)(\text{EBIT}) = 5,94,00,000$$

$$\text{EBIT} = \frac{5,94,00,000}{25} = ₹ 23,76,000$$

13. Capital Structure Theories



14. Basic Points

$$(A) K_d = \frac{\text{Interest}}{\text{Value of Debt}} = \frac{\text{Interest}}{V_d} \longrightarrow V_d = \frac{\text{Interest}}{K_d}$$

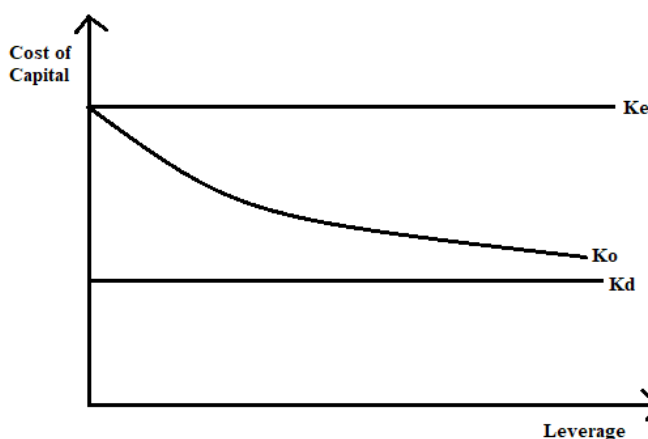
$$(B) K_e = \frac{EBIT - \text{Interest}}{\text{Value of Equity}} = \frac{EBT}{V_e} \longrightarrow V_e = \frac{EBT}{K_e}$$

$$(C) V_f = V_e + V_d$$

$$(D) K_o = \frac{EBIT}{(V_e + V_d)} = \frac{EBIT}{V_f} \longrightarrow V_f = \frac{EBIT}{K_o}$$

(E) If there is no debt then $K_o = K_e$

15. Net Income Approach



The optimum capital structure will be at a point where WACC is minimum and value of firm is maximum i.e. use maximum debt

- (1) Value of Debt = V_d ✓
- (2) K_d & K_e will be given and remain constant

$$(3) \text{ Value of Equity} = V_e = \frac{EBIT - \text{Interest}}{K_e}$$

$$(4) V_f = V_d + V_e$$

$$(5) K_o = \frac{EBIT}{V_f} = (K_e) \left(\frac{V_e}{V_f} \right) + (K_d) \left(\frac{V_d}{V_f} \right)$$

$$V_d = ₹. 50 \quad V_e = \frac{100 - (50 \times 10\%)}{15\%} = 333.333$$

$$V_f = 833.333$$

$$K_o = (15) \left(\frac{333.333}{833.333} \right) + (10) \left(\frac{50}{833.333} \right) = 12\%$$

$$V_d = 50 + 20 = 70 \quad V_e = \frac{100 - (70 \times 10\%)}{15\%} = 20$$

$$V_f = 70 + 20 = 90$$

$$K_o = (15) \left(\frac{20}{90} \right) + (10) \left(\frac{70}{90} \right) = 11.11\%$$

Question – 8

SK Ltd. has EBIT of ₹1,00,000. The company make use of debt and equity capital. The firm has 10% debentures of ₹5,00,000 and the firm's equity capitalization rate is 15%.

You are required to calculate:

- Current value of the firm ✓
- Overall cost of capital ✓
- Find value of firm and overall cost of capital if firm increases debt by ₹2,00,000

Solution

- (i) Calculation of total value of the firm

$$\text{Value of equity (S)} = \frac{\text{Earnings available for equity holders}}{K_e} = \frac{(1,00,000 - 50,000)}{0.15} = ₹3,33,333$$

$$\text{Value of Debt (D) (given)} = ₹5,00,000$$

$$\text{Total value of the firm (V)} = D + S = 5,00,000 + 3,33,333 = ₹ 8,33,333$$

- (ii) Overall cost of capital (K_o) = $K_e \left(\frac{S}{V} \right) + K_d \left(\frac{D}{V} \right) = 0.15 \left(\frac{₹3,33,333}{₹8,33,333} \right) + 0.10 \left(\frac{₹5,00,000}{₹8,33,333} \right) = 12.00\%$

$$\text{Or, } K_o = \frac{EBIT}{V} = \frac{₹1,00,000}{₹8,33,333} = 12.00\%$$

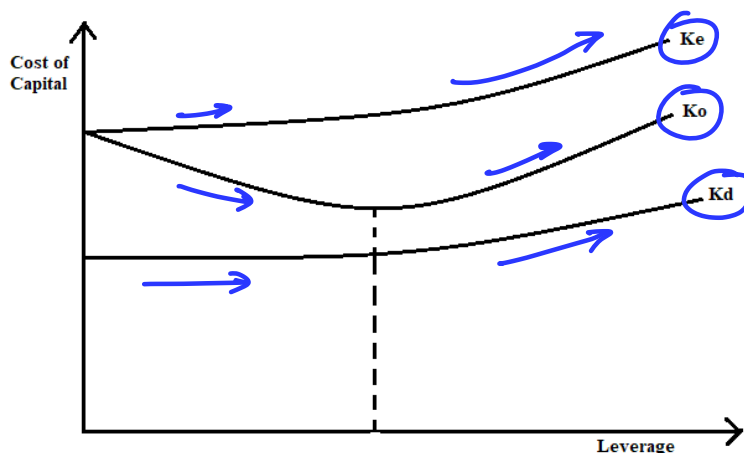
- (iii) Value of debt (D) = ₹5,00,000 + ₹2,00,000 = ₹7,00,000

$$\text{Value of equity (S)} = \frac{\text{Earnings available for equity holders}}{K_e} = \frac{(1,00,000 - 70,000)}{0.15} = ₹2,00,000$$

$$\text{Total value of the firm (V)} = D + S = 7,00,000 + 2,00,000 = ₹ 9,00,000$$

$$\text{Overall cost of capital (K}_o\text{)} = K_e \left(\frac{S}{V} \right) + K_d \left(\frac{D}{V} \right) = 0.15 \left(\frac{₹2,00,000}{₹9,00,000} \right) + 0.10 \left(\frac{₹7,00,000}{₹9,00,000} \right) = 11.11\%$$

$$\text{Or, } K_o = \frac{EBIT}{V} = \frac{₹1,00,000}{₹9,00,000} = 11.11\%$$

16. Traditional Approach

The optimum capital structure will be at a point where WACC is minimum and value of firm is maximum.

(1) Value of Debt = V_d

(2) K_d & K_e will be given and change for different levels

(3) Value of Equity = $V_e = \frac{EBIT - \text{Interest}}{K_e}$

(4) $V_f = V_d + V_e$

(5) $K_o = \frac{EBIT}{V_f} = (K_e) \left(\frac{V_e}{V_f} \right) + (K_d) \left(\frac{V_d}{V_f} \right)$

	Existing	30% Debt	50% Debt
V_d	—	$200 \times 30\% = 60$	$200 \times 50\% = 100$
V_e	$\frac{30-0}{16\%} = 18.75$	$\frac{30 - (60 \times 10\%)}{17\%} = 14.11765$	$\frac{30 - (100 \times 12\%)}{20\%} = 9$
V_f	18.75	20.11765	19
K_o ($EBIT \div V_f$)	$\frac{30}{18.75} =$	$\frac{30}{20.11765} =$	$\frac{30}{19} =$

Question – 9

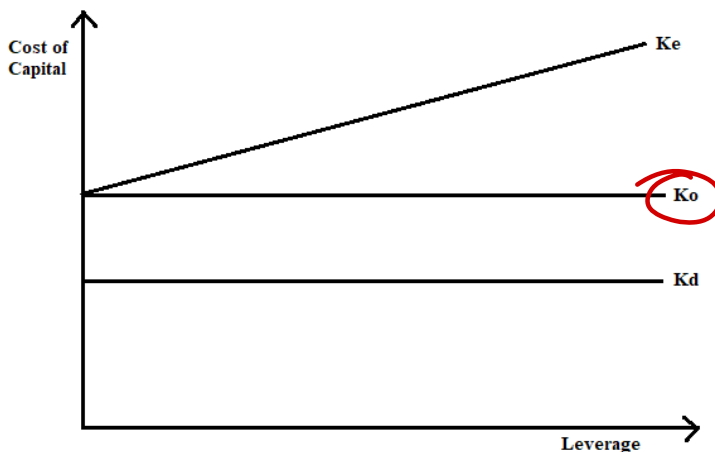
SK Ltd., is expecting an EBIT of ₹3,00,000. The company presently raised its entire fund requirement of ₹20 lakhs by issue of equity with equity capitalization rate of 16%. The firm is now contemplating to redeem a part of capital by introducing debt financing. The firm has two options to raise debt to the extent of 30% or 50% of total funds. It is expected that for debt financing upto 30% the rate of interest will be 10% and equity capitalization rate is expected to increase to 17%. However, if firm opts for 50% debt then interest rate will be 12% and equity capitalization rate will be 20%. You are required to compute value of firm and its overall cost of capital under different options if the traditional approach is held valid.

Solution

Statement of calculation of Value of Firm & K_o

Particulars	Existing	30% Debt	50% Debt
EBIT	3,00,000	3,00,000	3,00,000
(-) Interest	-	$6,00,000 \times 10\% = 60,000$	$10,00,000 \times 12\% = 1,20,000$
EAFE	3,00,000	2,40,000	1,80,000
Cost of Equity (K_e)	16%	17%	20%
Value of Equity (V_e) ($EAFE \div K_e$)	18,75,000	14,11,765	9,00,000
Value of Debt (V_d)	Nil	6,00,000	10,00,000
Value of Firm ($V_f = V_e + V_d$)	18,75,000	20,11,765	19,00,000
Cost of Capital (K_o) ($EBIT \div V_f$)	16%	14.91%	15.79%

17. Net Operating Income Approach



All options are optimum.

(1) Value of Debt = V_d

(2) K_d & K_o will be given and remain constant

(3) Value of Firm = $V_f = \frac{EBIT}{K_o}$

(4) $V_e = V_f - V_d$

(5) $K_e = \frac{EBIT - \text{Interest}}{V_e}$

$$V_{FA} = \frac{4.502}{18\%} = 252 \quad \begin{cases} V_E = 252 \times 40\% = 102 \\ V_D = 252 \times 60\% = 150 \end{cases}$$

$$Mo. \times Inc. = 3\% \times [4.502 - (152 \times 8\%)] = 9900$$

$$V_{FB} = \frac{4.502}{18\%} = 252 \quad \begin{cases} V_E = 252 \times 80\% = 202 \\ V_D = 252 \times 20\% = 50 \end{cases}$$

$$K_{EB} = \frac{4.502 - (50 \times 8\%)}{202} = 20.5\%$$

Question – 10

A Limited and B Limited are identical except for capital structures. A Ltd. has 60% debt and 40% equity, whereas B Ltd. has 20% debt and 80% equity. (All percentages are in market-value terms). The borrowing rate for both companies is 8% in a no-tax world, and capital markets are assumed to be perfect.

(a) (i) If X owns 3% of the equity shares of A Ltd. determine his return if the company has net operating income of ₹ 4,50,000 and the overall capitalization rate of the company, (K_o) is 18%.

(ii) Calculate the implied required rate of return on equity of A Ltd.

$$K_e = \frac{4.502 - (152 \times 8\%)}{(252 - 152)} = 33\%$$

(b) B Ltd. has the same net operating income as A Ltd.

(i) Calculate the implied required equity return of B Ltd.

(ii) Analyze why does it differ from that of A Ltd.

Solution

(a) (i) Value of A Ltd. = $\frac{EBIT}{K_o} = \frac{4,50,000}{18\%} = ₹ 25,00,000$

Value of Debt = ₹ 25,00,000 × 60% = ₹ 15,00,000

Value of Equity = ₹ 25,00,000 × 40% = ₹ 10,00,000

Income Statement

EBIT	4,50,000
Less: <u>Interest (15,00,000 × 8%)</u>	<u>1,20,000</u>
EBT / EAT / EAE	<u>3,30,000</u>
Return on 3% shares of Mr. X = ₹ 3,30,000 × 3% = ₹ 9,900	

$$(ii) \text{ Implied rate of return on equity} = \frac{EAE}{\text{Value of equity}} \times 100 = \frac{3,30,000}{10,00,000} \times 100 = 33\%$$

- (b) (i) Value of B Ltd. = $\frac{EBIT}{K_e} = \frac{4,50,000}{18\%} = ₹ 25,00,000$
 Value of debt = ₹ 25,00,000 × 20% = ₹ 5,00,000
 Value of equity = ₹ 25,00,000 × 80% = ₹ 20,00,000

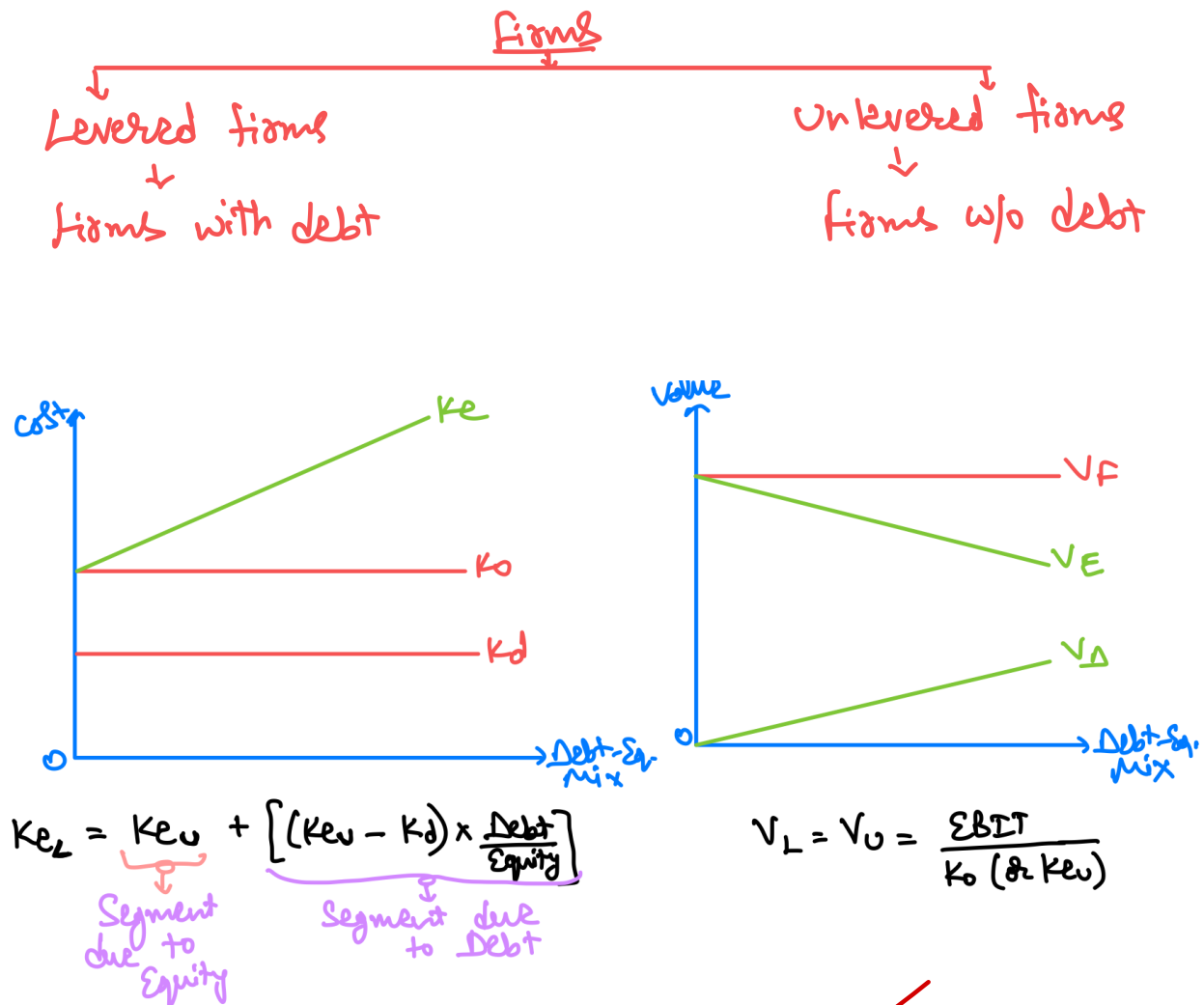
Income Statement

EBIT	4,50,000
Less: <u>Interest (5,00,000 × 8%)</u>	<u>40,000</u>
EBT / EAT / EAE	<u>4,10,000</u>

$$\text{Implied rate of return on equity} = \frac{EAE}{\text{Value of equity}} \times 100 = \frac{4,10,000}{20,00,000} \times 100 = 20.50\%$$

- (ii) It is lower than the A Ltd. because B Ltd. uses less debt in its capital structure. As the equity capitalization is a linear function of the debt-to-equity ratio when we use the net operating income approach, the decline in required equity return offsets exactly the disadvantage of not employing so much in the way of “cheaper” debt funds.

18. MM Model without Tax



$$K_0 = \frac{\text{EBIT}}{V_F} = \frac{21,600}{1200} = 18\% = K_{eU}$$

Question – 11

SK Ltd. has a net operating income of ₹ 21,60,000 and the total capitalization of ₹ 120 lakhs. The company is evaluating the options to introduce debt financing in the capital structure and the following information is available at various levels of debt value.

$$K_{eL} = K_{eU} + (K_{eU} - K_d) \left(\frac{V_D}{V_U} \right)$$

Debt Value (₹) (V_D)	Interest rate (%)	Equity Capitalization rate (%)
0	NA	12.00
10,00,000	7.00	12.50
20,00,000	7.00	13.00
30,00,000	7.50	13.50
40,00,000	7.50	14.00
50,00,000	8.00	15.00
60,00,000	8.50	16.00
70,00,000	9.00	17.00
80,00,000	10.00	20.00

$$18 + (18 - 0) \left(\frac{0}{1200} \right) = 18\%$$

$$18 + (18 - 7) \left(\frac{100}{1200} \right) = 19\%$$

$$18 + (18 - 7) \left(\frac{200}{1200} \right) = 20.2\%$$

You are required to compute the equity capitalization rate if MM approach is followed. Assume that the firm operates in zero tax regime and calculations to be based on book values.

Solution

As per MM approach, cost of the capital (K_0) remains constant, and cost of equity increases linearly with debt.

$$\text{Value of a Firm} = \frac{\text{NOI}}{K_0}$$

$$\therefore 1,20,00,000 = \frac{21,60,000}{K_0}$$

$$\therefore K_0 = \frac{21,60,000}{1,20,00,000} = 18\%$$

Under MM approach, $k_e = k_0 + \frac{D}{E}(k_0 - k_d)$

Statement of equity capitalization under MM approach

Debt Value (₹)	Equity Value (₹)	Debt/Equity	K_d (%)	K_0 (%)	$(K_0 - K_d)$	$K_e = K_0 + (K_0 - K_d)\left(\frac{D}{E}\right)$
-	1,20,00,000	0.0000	NA	18.00	18.00	18.00 ✓
10,00,000	1,10,00,000	0.0909	7.00	18.00	11.00	19.00 ✓
20,00,000	1,00,00,000	0.2000	7.00	18.00	11.00	20.20 ✓
30,00,000	90,00,000	0.3333	7.50	18.00	10.50	21.50
40,00,000	80,00,000	0.5000	7.50	18.00	10.50	23.25
50,00,000	70,00,000	0.7143	8.00	18.00	10.00	25.14
60,00,000	60,00,000	1.0000	8.50	18.00	9.50	27.50
70,00,000	50,00,000	1.4000	9.00	18.00	9.00	30.60
80,00,000	40,00,000	2.0000	10.00	18.00	8.00	34.00

$$K_0 = (K_e)(w_e) + (K_d)(w_d) \\ 16 = (K_e)\left(\frac{2}{3}\right) + (10)\left(\frac{1}{3}\right) \Rightarrow 16 = \frac{2K_e + 10}{3} \Rightarrow 48 = 2K_e + 10 \Rightarrow K_e = 19\%$$

Question – 12

One-third of the total market value of SK Ltd. consists of loan stock, which has a cost of 10 percent. Another company MK Ltd., is identical in every respect to SK Ltd., except that its capital structure is all-equity, and its cost of equity is 16 percent. According to Modigliani and Miller, if we ignored taxation and tax relief on debt capital, compute the cost of equity of SK Ltd.

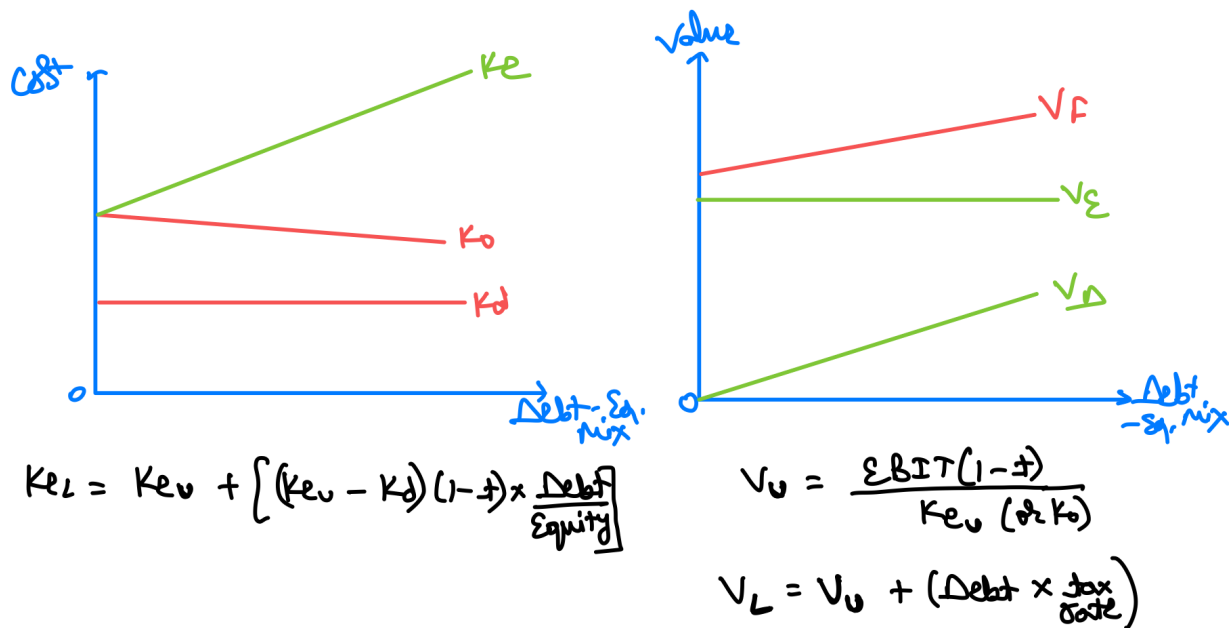
Solution

Cost of equity in SK Ltd. (levered company) will be calculated as follows:

$$K_0 = \frac{2}{3} \times K_e + \frac{1}{3} \times K_d = 16\% \text{ (i.e. equal to WACC of SK Ltd.)}$$

$$\text{Or, } 16\% = \frac{2}{3} \times K_e + \frac{1}{3} \times 10\% \text{ Or, } K_e = 19\%$$

19. MM Model with Tax



Question – 13

The following data relate to two companies belonging to the same risk class:

Particulars	A Ltd.	B Ltd.
Expected Net Operating Income	→ ₹ 18,00,000	→ ₹ 18,00,000
12% Debt	→ ₹ 54,00,000	→ 0
Equity Capitalization Rate	-	18% → $K_{eU} = K_o$

Required:

- Determine the total market value, equity capitalization rate and weighted average cost of capital for each company assuming no taxes as per MM approach
- Determine the total market value, equity capitalization rate and weighted average cost of capital for each company assuming 40% taxes as per MM approach.

Solution

$$(c) \text{ Value of B Ltd. (Unlevered firm)} = \frac{\text{EBIT}}{K_e} = \frac{18,00,000}{18\%} = ₹ 1,00,00,000$$

$$\begin{aligned} \text{Value of A Ltd. (Levered firm)} &= \text{Value of B Ltd.} + \text{Tax benefit} \\ &= 1,00,00,000 + (54,00,000 \times 0) = ₹ 1,00,00,000 \end{aligned}$$

Ke of B Ltd. = 18% (given)

$$K_e \text{ of A Ltd.} = \frac{\text{EBIT} - \text{Interest}}{\text{Value of Equity}} = \frac{18,00,000 - (54,00,000 \times 12\%)}{1,00,00,000 - 54,00,000} = \frac{11,52,000}{46,00,000} = 0.2504 = 25.04\%$$

WACC of B Ltd. = Ke = 18%

WACC of A Ltd.

Source (1)	Amount (2)	Weights (3)	Cost of capital (4)	Weighted Average Cost (5)= (3)x(4)
Equity	46,00,000	0.46	25.04	11.52
Debt	54,00,000	0.54	12.00	6.48
		1		18

Weighted Average Cost of Capital (WACC) = 18%

$$(d) \text{ Value of B Ltd. (Unlevered firm)} = \frac{EBIT(1-t)}{K_e} = \frac{18,00,000(1-0.40)}{18\%} = ₹ 60,00,000$$

$$\begin{aligned} \text{Value of A Ltd. (Levered firm)} &= \text{Value of B Ltd.} + \text{Tax benefit} \\ &= 60,00,000 + (54,00,000 \times 0.40) = ₹ 81,60,000 \end{aligned}$$

K_e of B Ltd. = 18% (given)

$$K_e \text{ of A Ltd.} = \frac{(EBIT - \text{Interest})(1-t)}{\text{Value of Equity}} = \frac{[18,00,000 - (54,00,000 \times 12\%)](1-0.40)}{81,60,000 - 54,00,000} = \frac{6,91,200}{27,60,000} = 25.04\%$$

WACC of B Ltd. = K_e = 18%

$$K_d \text{ of A Ltd.} = I \times (1-t) = 12 \times (1-0.40) = 7.20\%$$

WACC of A Ltd.

Source (1)	Amount (2)	Weights (3)	Cost of capital (4)	Weighted Average Cost (5)= (3)x(4)
Equity	27,60,000	0.34	25.04	8.51
Debt	54,00,000	0.66	7.20	4.75
		1		13.26

Weighted Average Cost of Capital (WACC) = 13.26%

Question – 14

Rounak Ltd. is an all equity financed company with a market value of ₹ 25,00,000 and cost of equity (K_e) 21%. The company wants to buyback equity shares worth ₹ 5,00,000 by issuing and raising 15% perpetual debt of the same amount. Rate of tax may be taken as 30%. After the capital restructuring and applying MM model (with taxes), you are required to compute:

- ✓(a) Market value of the company
- ✓(b) Cost of equity
- ✓(c) Weighted average cost of capital (using market weights) and comment on it.

Solution**Working Note:**

$$\text{Market value of equity} = \frac{\text{Net Income (NI) for Equity Holders}}{K_e}$$

$$₹ 25,00,000 = \frac{\text{Net Income (NI) for Equity Holders}}{0.21}$$

$$\text{Net Income for Equity Holders} = 25,00,000 \times 0.21 = ₹ 5,25,000$$

$$EBIT = \frac{5,25,000}{1-0.30} = ₹ 7,50,000$$

(₹ in lakhs)

Particulars	All Equity	Debt and Equity
EBIT	7,50,000	7,50,000
(-) Interest	-	(75,000)
EBT	7,50,000	6,75,000
(-) Tax @ 30%	2,25,000	2,02,500
Income to shareholders	5,25,000	4,72,500

- (a) Market value of company = Value of equity + Value of debt
 $= ₹ 25,00,000 + (5,00,000 \times 0.30) = ₹ 26,50,000$
 The impact is that the market value of the company has increased by ₹ 1,50,000.

(b) $K_e = \frac{\text{Net income to equity holders}}{\text{Equity value}} = \frac{4,72,500}{26,50,000 - 5,00,000} = 0.219 = 21.98\%$

(c) $K_d = I \times (1 - t) = 15\% \times (1 - 0.30) = 10.5\%$

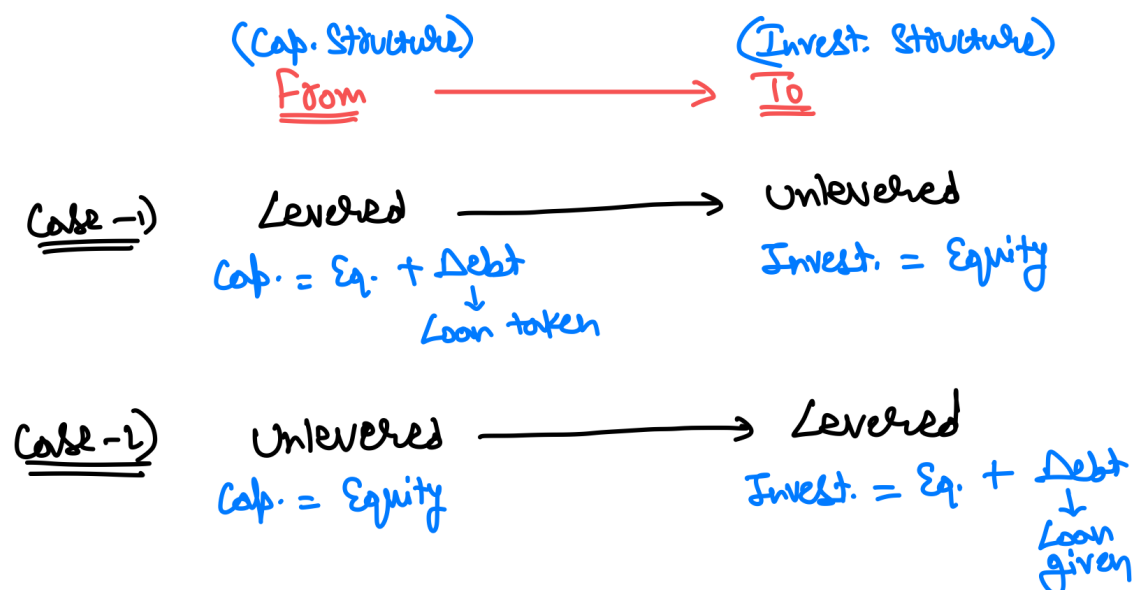
Weighted Average Cost of Capital (WACC)

Source (1)	Amount (2)	Weights (3)	Cost of capital (4)	Weighted Average Cost (5) = (3) x (4)
Equity	21,50,000	0.81	21.98	17.80
Debt	5,00,000	0.19	10.50	2.00
	26,50,000	1		19.80

Weighted Average Cost of Capital (WACC) = 19.80%

The impact is that WACC has fallen by 1.20% due to benefit of lower cost of capital of debt.

20. Arbitrage Process



Question – 15

The following is the data regarding two companies S and K belonging to the same risk class:

		<u>Company S</u>	<u>Company K</u>
Number of ordinary shares	→	90,000	1,50,000
Market price per share (₹)	→	1.20	1.00
6% Debentures (₹)	→	60,000	--
Profit before interest (₹) (EBIT)	→	18,000	18,000

All profits after debenture interest are distributed as dividends. Explain how under Modigliani & Miller approach, an investor holding 10% shares in company S will be better off in switching his holding to Company K.

Solution

Particulars	Company S	Company K
Value of Equity	$90,000 \times 1.20 = 1,08,000$	$1,50,000 \times 1 = 1,50,000$
Value of Debt	60,000	-
Total value of Firm	1,68,000	1,50,000

Value of levered company S is more than unlevered company therefore investor will sell his shares in Company S and buy shares of Company K. To maintain the risk level i.e. Debt & equity ratio, he will borrow proportionate amount and invest that in shares of company K.

Investment & Borrowings:

Sell value from shares of Company S ($1,08,000 \times 10\%$)	₹ 10,800
Borrow money ($60,000 \times 10\%$)	₹ 6,000
Buy shares of Company K	₹ 16,800

Earning of Investor

Income from shares of Company K $\left(\frac{18,000}{1,50,000} \times 16,800\right)$	₹ 2,016
Less: Interest on loan ($6,000 \times 6\%$)	₹ 360
Net income from Company K	₹ 1,656
Less: Income from Company S $[(18,000 - 3,600) \times 10\%]$	₹ 1,440
Incremental gain due to arbitrage	₹ 216

Question – 16

The following data relate to two companies belonging to the same risk class:

Particulars	S Ltd.	K Ltd.
Expected Net Operating Income	₹ 2,40,000	₹ 2,40,000
10% Debt	₹ 7,20,000	-
Equity Capitalization Rate	20%	15%

Required:

- Determine the total value and the weighted average cost of capital for each company assuming no taxes before the start of arbitrage process.
- Show the arbitrage process by which an investor who holds 10% equity share in K Ltd. will be benefited by investing in S Ltd.
- When will this arbitrage process come to an end?

Solution

(a) Statement of calculation of value of firm

Particulars	Company S	Company K
EBIT	2,40,000	2,40,000
(-) Interest	72,000	-
EBT	1,68,000	2,40,000
Ke	20%	15%
Ve	$\frac{1,68,000}{20\%} = 8,40,000$	$\frac{2,40,000}{15\%} = 16,00,000$
Vd	7,20,000	-
Vf	15,60,000	16,00,000
Ko	$\frac{2,40,000}{15,60,000} \times 100 = 15.385\%$	$\frac{2,40,000}{16,00,000} \times 100 = 15\%$

(b) Value of Company K (unlevered) is more than of Company S (Levered). Thus, investor will sell shares in Company K and buy shares of Company S. To maintain the level of risk i.e. Debt and Equity ratio (7.2 : 8.4), he will lend proportionate amount and invest balance amount in shares of company K.

Investment & Borrowings:

Sale value from shares of Company K ($16,00,000 \times 10\%$)	<u>₹ 1,60,000</u>
Lend money [$1,60,000 \times (7.2 \div 15.6)$]	₹ 73,846
Buy shares of Company S [$1,60,000 \times (8.4 \div 15.6)$]	<u>₹ 86,154</u>
	<u>₹ 1,60,000</u>

Earning of Investor

Income from shares of Company S $\left(\frac{1,68,000}{8,40,000} \times 86,154\right)$	₹ 17,231
Add: Interest from loan given $(73,846 \times 10\%)$	₹ 7,385
Net income from Company S	₹ 24,616
Less: Income from Company K $(2,40,000 \times 10\%)$	₹ 24,000
Incremental gain due to arbitrage	₹ 616

(d) The arbitrage process will come to an end when the value of both firms i.e. S and K becomes equal.

Question – 17

Company P and Q are identical in all respects including risk factors except for debt/equity, company P having issued 10% debentures of ₹ 18 lakhs while company Q is unlevered. Both the companies earn 20% before interest and taxes on their total assets of ₹ 30 lakhs. $\rightarrow EBIT = 30 \times 20\% = 60$

Assuming a tax rate of 50% and capitalization rate of 15% from an all-equity company.

Required to calculate the value of companies P and Q using (a) Net Income Approach and (b) net Operating Income Approach.

$\rightarrow$ MM with tax $\rightarrow V_Q = \frac{60(1-0.50)}{15\%} = 200$ $V_P = 200 + (18 \times 50\%) = 290$

Solution**(a) Valuation under Net Income Approach**

Particulars	P (₹)	Q (₹)
EBIT $(30,00,000 \times 20\%)$	6,00,000	6,00,000
Less: Interest $(18,00,000 \times 10\%)$	1,80,000	-
EBT	4,20,000	6,00,000
Less: Tax @ 50%	2,10,000	3,00,000
EAT/EAE	2,10,000	3,00,000
Value of Equity $(V_e = EAE \div 15\%)$	$\frac{2.100}{15\%} = 14,00,000$	$\frac{30}{15\%} = 20,00,000$
Add: Total value of debt (V_d)	18,00,000	-
Total value of company $(V_e + V_d)$	32,00,000	20,00,000

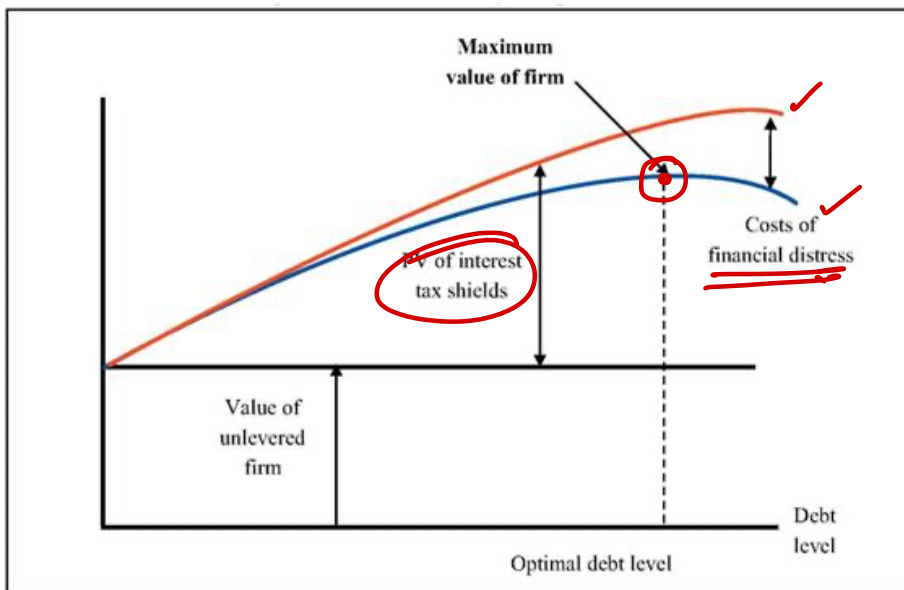
NI
 $\begin{matrix} K_e \\ \hline P_d \end{matrix}$

(b) Valuation under Net Operating Income Approach

$$\text{Value of Firm Q (unlevered)} = \frac{EBIT(1-t)}{K_e} = \frac{6,00,000 \times (1-0.50)}{15\%} = ₹ 20,00,000$$

$$\begin{aligned} \text{Value of Firm P (levered)} &= \text{Value of unlevered firm} + (\text{Debt} \times \text{Tax rate}) \\ &= 20,00,000 + (18,00,000 \times 50\%) = ₹ 29,00,000 \end{aligned}$$

21. Trade-Off Theory



22. Pecking Order Theory

